

Full-Stack Architecture Overview (1-Page Note)

Full-stack architecture refers to the complete structure of a web application, including both the **frontend (client-side)** and **backend (server-side)** components. It represents how data flows through the system, how users interact with the app, and how the server processes and stores information.

1. Client-Side (Frontend)

The frontend is what the user sees and interacts with in a browser or mobile app. It is responsible for delivering a smooth, responsive, and visually appealing user interface.

Key Technologies:

- **HTML** – Structure of web pages
- **CSS** – Visual styling and layout
- **JavaScript** – Interactivity and browser logic
- **Frontend Frameworks** – React, Angular, Vue

Responsibilities:

- Displaying content
 - Handling user input
 - Communicating with backend APIs
 - Managing UI state
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2. Server-Side (Backend)

The backend handles logic, processing, authentication, and communication with the database.

It receives requests from the frontend and returns responses.

Key Technologies:

- **Python (Django / Flask)**
- **Node.js**
- **Java / Spring Boot**
- **PHP / Laravel**

Responsibilities:

- User authentication
 - Business logic
 - Data validation
 - Handling API requests
 - Serving responses to the frontend
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3. Database Layer

The database stores and manages data used by the application.

Database Types:

- **SQL:** MySQL, PostgreSQL, SQLite (structured tables)
- **NoSQL:** MongoDB, Firebase (document/key-value storage)

Responsibilities:

- Storing user data
 - Fetching, updating, deleting records
 - Maintaining data integrity
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4. APIs (Application Programming Interfaces)

APIs connect the frontend and backend, allowing them to communicate smoothly.

Common Types:

- **REST APIs**
- **GraphQL APIs**

Responsibilities:

- Sending data from frontend → backend
 - Returning processed data (JSON)
 - Handling HTTP methods (GET, POST, PUT, DELETE)
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5. Full-Stack Data Flow (Simple View)

1. User interacts with **frontend UI**

2. Frontend sends request to **backend API**
 3. Backend processes logic and interacts with **database**
 4. Database returns data to backend
 5. Backend sends response to frontend
 6. UI updates based on returned data
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6. Deployment Layer

Full-stack apps are deployed on servers or cloud platforms.

Popular Deployment Platforms:

- AWS
 - Render
 - Railway
 - Heroku
 - Vercel
 - PythonAnywhere
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Summary

Full-stack architecture combines both **frontend + backend + database** with smooth communication through APIs. A full-stack developer understands how all these layers work together to build complete, scalable, user-friendly applications.